

$$\begin{array}{ccc} \frac{17}{A^{[0]}} & \frac{3}{Z^{[1]} \quad A^{[1]}} & \frac{1}{Z^{[2]} \quad A^{[2]}} \end{array}$$

$$A^{[0]} = X \quad (l \times 17)$$

$$Z^{[1]} = \text{net} \left(\begin{array}{cc} A^{[0]} & W^{[1]} \\ l \times 17 & 17 \times 3 \end{array} \right) + b^{[1]} \quad 1 \times 3$$

$$A^{[1]} = R(A^{[1]}) \quad l \times 3$$

broadcasted:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} + [1 \ 1 \ 1] = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix}$$

$$Z^{[2]} = \text{net} \left(\begin{array}{cc} A^{[1]} & W^{[2]} \\ l \times 3 & 3 \times 1 \end{array} \right) + b^{[2]} \quad 1 \times 1$$

$$A^{[2]} = \text{Sigmoid}(A^{[2]}) \quad l \times 1$$

OUTPUT
LAYER
 $l \times \#out$

for use

$$\delta_{\text{OUTPUT}} = (y - A^{[2]}) f'_{(-1)}(z^{[2]})$$

$l \times 1 \quad l \times 1 \quad l \times 1$

$$\delta_{\text{PREV LAYER}} = A^{[1]}$$

$l \times 3$

$$\Delta W^{[2]} = \text{dot} \left(A^{[1]T}, \delta_t \right)$$

$3 \times l \quad l \times 1$

$$\delta_t o_v \quad (o_v \cdot \delta_t)$$

$$\Delta b^{[2]} = \delta_{\text{OUTPUT}} \cdot \text{SUM}(\text{Axis} = 0)$$

$1 \times 1 \quad l \times 1 \rightarrow 1 \times 1$ (same size pattern)

HIDDEN
LAYER

$$\delta_j = \sum_{k=1}^K \delta_k \cdot w_{kj} f'_k(\text{net}_k)$$

$$\delta^{[1]} = \left(\delta_{\text{OUTPUT}} \cdot W^{[2]T} \right) f'_{(-1)}(z^{[2]})$$

$l \times 3 \quad l \times 1 \quad 1 \times 3 \quad l \times 1$

$l \times 3$

broadcasted, connections

$$\Delta W^{[1]} = \text{dot} \left(A^{[0]T}, \delta^{[1]} \right)$$

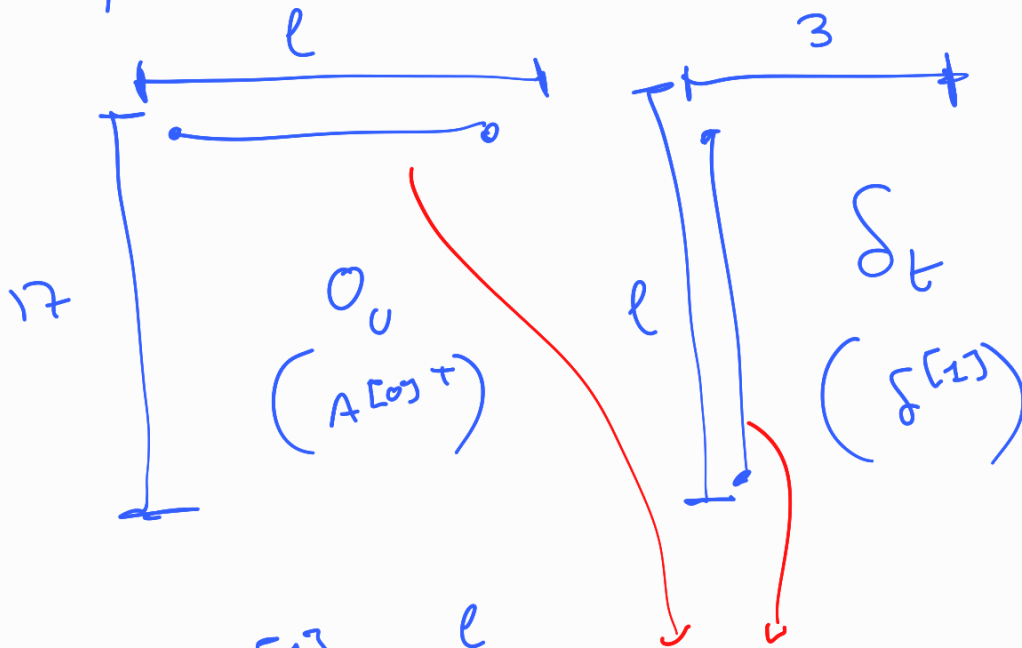
$17 \times l \quad l \times 3$

$$\delta_t o_v \quad (o_v \cdot \delta_t)$$

$$\Delta b^{[1]}_{1 \times 3} = \sum_{l \times 3 \dots \dots 1 \times 3} (S^{[1]})$$

why dot?

ovviamente
tutto / l ...



$$\Delta x^{[1]} = \sum_{p=1}^l O_v \cdot \delta_t$$

in single
element